

The Pass Rush: Moves, Stunts, and Why Pressure Beats Sacks

Four men, about two and a half seconds, and a 315-pound obstacle each: how a defense gets to the quarterback without blitzing, and how to see it from the couch

Third-and-8, early in the second quarter. The offense spreads four receivers and keeps the back in to block; the defense shows four linemen and nobody else near the line. At the snap the right defensive end takes three long strides up the outside, and the offensive tackle in front of him kick-slides back fast to stay between him and the quarterback. Then, with the tackle's weight going backward, the end plants his outside foot and drives straight into the tackle's chest. The tackle's feet stop moving; he is walked backward three yards, into the spot where the quarterback wanted to set his feet. The quarterback throws a beat early, off his back foot, and the pass sails over a receiver who was about to come open. Incomplete. Punt. The box score records nothing for the end: no sack, no tackle, not even a hit. His position coach will grade it as one of his best plays of the day. This chapter is about why.

You'll be able to:

- Name the main pass-rush moves (speed, bull, long arm, swim, rip, spin) as you see them, and say which mistake by the blocker each one punishes.
- Explain how a rusher *plans* his moves across a game, and spot the counter when a tackle starts guessing.
- Recognise a stunt (T-E, E-T, interior twist) from the crossing paths, and say what it was trying to do to the protection.
- Read rush lanes and contain, and explain why defenses "mush rush" a quarterback who can run, and what that costs them.
- Explain why pressure is a steadier measure of a pass rush than sacks, and how much of a sack belongs to the quarterback.
- Use pass rush win rate, get-off and time to pressure, and know what each can and cannot tell you.

You'll need

- **Pass Protection:** the pocket is a cup (the interior holds, the tackles give ground); slide protection blocks gaps and man protection blocks men; the tackle's kick slide and vertical set, the blocker's punch and anchor; linemen passing off a stunt; and the numbers: pressure, pressure rate, time to throw. This chapter is the same fight from the other side.
- **Defensive Line Play:** technique numbers from the defense's side (the **3-technique** on the guard's outside shoulder, the **1** shaded on the center, the **5** and the wide-**9** outside the tackle), one-gap penetration, and contain: edge rushers not running past the quarterback's depth.

How to read the diagrams. Offense is blue circles, defense orange squares, and *left* and *right* are the offense's. The linemen are LT, LG, C, RG and RT; Q is the quarterback (in most diagrams drawn at his launch point, the spot he throws from) and R the running back. On defense, E is an end and the tackles are labelled by technique (**1**, **3**); M is a linebacker. An orange arrow is a rusher's path; a blue line ending in a short bar is a blocker's set; a dashed dark-blue line ending in a bar is a blocker being driven backward. A thick dark-orange stroke ending in a dot is the rusher's arm or hand at the moment of a move, and a curved dark-orange arrow is a hand move (a swim over the top, a rip up through). A faint, dashed square is where a rusher is at a moment the diagram does not otherwise show. Shaded orange areas are rush lanes; red marks a free path or an escape lane.

The rusher's problem: a race and a fight

Start with what a pass rusher is up against, because every move in this chapter is an answer to it.

The quarterback in today's NFL releases the ball a little under three seconds after the snap on average (Pass Protection has the numbers), and on a quick throw in well under two. An edge rusher who lines up outside the tackle has to cover an arc of about thirteen yards to reach the spot where the quarterback sets his feet; a defensive tackle in the middle has nine or so, straight ahead, but through the thickest traffic on the field. Unblocked, an edge rusher would cover his arc in roughly two seconds. He is never unblocked. A man who weighs about 315 pounds, who knows where the quarterback will be and only has to stay in the way, is standing on the path.

So a pass rush is two contests at once. It is a **race**: the rusher has to get to the quarterback before the ball leaves, and a rusher who arrives at 3.4 seconds against a quarterback who threw at 2.6 has done nothing, however good his move was. And it is a **fight**: to win the race he has to beat a blocker who is bigger, who has the inside position, and who is allowed to use his hands too. The

rusher's advantages are the two the blocker lacks. He knows the snap is coming and where he is going, and he moves *forward*, while the blocker moves backward.

There is a third fact, and it explains why defenses pay pass rushers so much: **most of the time, the defense rushes only four**. In FTN's charting of the 2023–2025 regular seasons, 69% of dropbacks faced exactly four rushers and 26% faced five or more.¹ A blitz can manufacture a free rusher by sending more men than the offense can block, but every extra rusher is one fewer defender in coverage, and the quarterback knows it. Four rushers against five or six blockers leaves seven defenders to cover at most five receivers. If those four can create pressure by themselves, the defense gets to the quarterback *and* covers everybody. That is the holy grail of defensive football, and this chapter is about how it is done: by moves, by plans, and by stunts.

! Why it exists

A four-man pass rush exists because coverage is expensive and pressure is valuable. Each rusher the defense adds is a defender it takes out of coverage; each rusher it subtracts gives the quarterback more time. The four-man rush is the balance point, and a team whose four can win their one-on-one fights gets pressure without paying for it in coverage. That is why an edge rusher who wins those fights is usually the highest-paid player on his defense, and why everything else in this chapter, from rush moves to stunts to the mush rush, is an attempt to win with four.

Get-off: the first step decides more than you think

The fight begins before anyone touches anyone. A rusher's **get-off** is his first movement at the snap: how quickly he reacts to the ball moving and gets across the line of scrimmage.

Defenders are not allowed to move into the neutral zone before the snap, and unlike the offense they do not know the snap count. So a rusher does not listen to the quarterback's cadence; he watches the ball, or the center's hand on it, and goes the instant it moves. The offense's weapons against this are a varied snap count (a "hard count" to draw a rusher offside, a quick count to catch him flat-footed) and, in a loud stadium, a silent count that gives nothing away. The rusher's weapon is a stance with his weight already tilted forward and a first step that covers ground without popping up.

Why does a tenth of a second matter so much? Because of what it does to the meeting point.

¹Author's calculation via nflreadpy: the participation file's `number_of_pass_rushers` (FTN charting) on regular-season dropbacks, 2023–2025. Pass Protection gives the same split for 2024–2025.

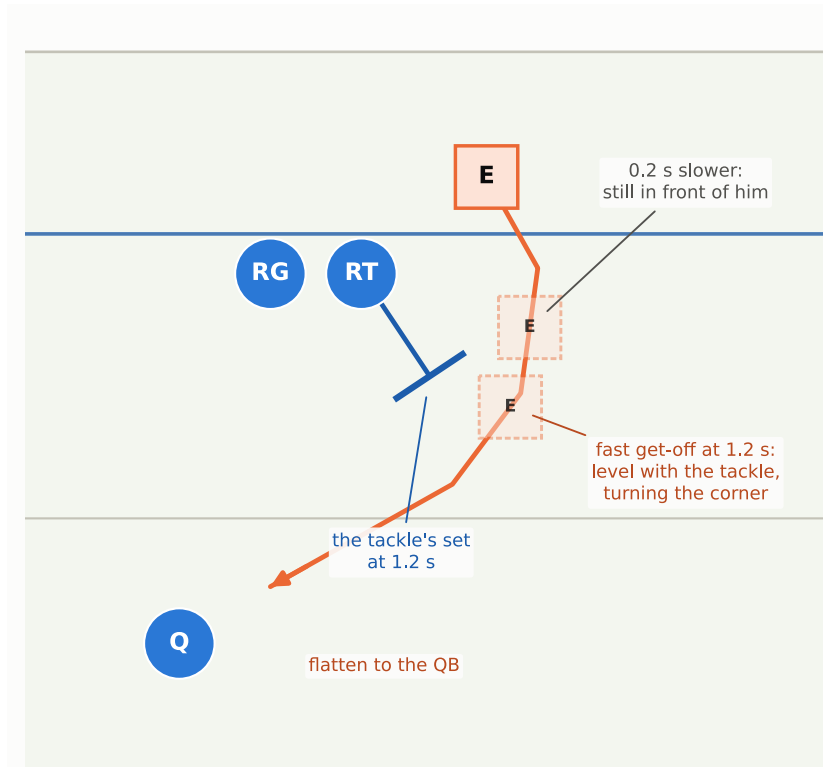


Figure 1: Two edge rushers on the same arc against the same tackle, 1.2 seconds after the snap (faint squares; the line is the path both run). The tackle has kick-slid back to his set point (bar). The rusher who moved the instant the ball did (fast get-off) has already reached the tackle's depth and is turning the corner, his hips level with the tackle's: the tackle has to open up and run with him, and is in trouble. The rusher who started two-tenths of a second later is still more than a yard in front of the tackle, where the tackle can meet him square and punch. Same speed, same path; the only difference is the first step.

At full speed an edge rusher covers about seven yards a second, so a tenth of a second at the snap is most of a yard on the arc. In Figure 1 the two rushers differ by two-tenths, and by 1.2 seconds that has become a 1.5-yard difference in where they meet the tackle. The quick one meets him *level*, at the tackle's own depth, where the tackle has to turn his hips to run with him, and a blocker who turns his hips has given up the inside and lost most of his power. The slow one meets him *in front*, where the tackle is square, balanced and ready to punch. Same athlete, same path, a completely different fight.

That is why scouts time rushers' first steps at the Combine, why defensive line coaches drill get-off every single day, and why the best rushers are often the first player on the field to move. Tracking data now measures it directly: the league's Next Gen Stats measures a rusher's get-off time, how long he takes to cross the line of scrimmage after the snap, and quotes it in its broadcasts and analysis (the fastest are around a third of a second).² You will compute one from simulated tracking data at the end of this chapter.

The arc and the corner

Before the moves, the geometry they are all trying to bend. An edge rusher's basic path is the **arc**: a curve up the outside of the tackle and around him toward the quarterback. The tackle's answer, as Pass Protection showed, is to kick-slide back and out, staying between the rusher and the quarterback, and to push the rusher *past* the quarterback if he keeps running. The pocket is a cup, and a rusher run around the rim of the cup ends up behind it, harmless.

The rusher wins the arc at one spot: the **corner**, the moment his hips get level with the tackle's. Before that, the tackle is in front of him and can punch and steer. After it, the tackle is beside or behind him, can only push, and any push sends the rusher where he wanted to go. That is why the last part of every good edge rush is a tight turn back toward the quarterback, **flattening** the arc once he is past the tackle rather than continuing upfield.

²Next Gen Stats Analytics Team, "Next Gen Stats: Introduction to Pressure Probability," NFL.com (2023), which measures get-off as the time a rusher takes to cross the line of scrimmage after the snap and cites a 0.33-second get-off by Micah Parsons as one of the fastest of the season: <https://www.nfl.com/news/next-gen-stats-introduction-to-pressure-probability>.

Two physical skills decide whether a rusher can do this, and you can see both on any broadcast replay. The first is **bend**: the ability to lean his whole body toward the inside, ankles flexed, shoulder almost at the turf, while running at full speed, so he can turn the corner tightly instead of drifting wide. The second is **dip**: dropping the shoulder nearest the tackle so the tackle's hands find nothing to grab. Watch a great edge rusher win and you will see a body tilted at an angle that looks like it should fall over.

The tackle, of course, knows all this. If he gets deep enough fast enough, there is no corner to turn. And so the rusher needs other ways to win: through the tackle, or back inside him. That is what the moves are.

The moves

There are dozens of named pass-rush moves, and every rusher has his own blend, but almost all of them belong to a handful of families. Six are worth knowing by sight. In each panel of Figure 2 the rusher (E) lines up outside the right tackle; read the note under each panel for what the move attacks.

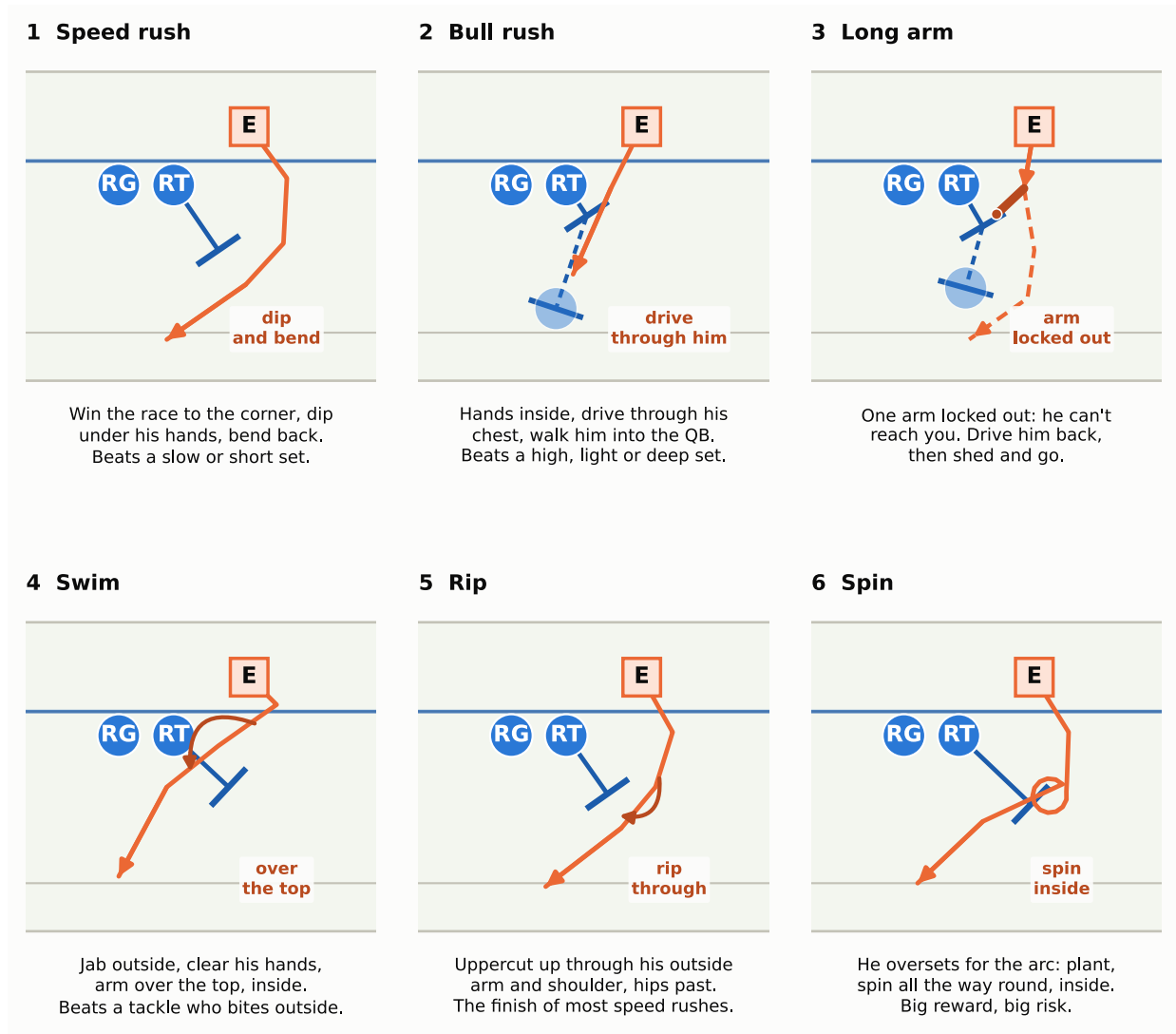


Figure 2: Six pass-rush moves against a right tackle (zoomed in on him; the quarterback is off the bottom-left of each panel, where the rush arrows are heading). Speed rush: beat him to the corner and bend back. Bull rush: straight through his chest, walking him backward (dashed). Long arm: one arm locked into his chest, so he cannot reach the rusher, then drive and escape. Swim: sell outside, then throw the arm over the top and go inside. Rip: an uppercut through his outside arm to get the hips past. Spin: when he has overcommitted to the outside, spin back inside. Each move beats a different mistake; the rusher's art is choosing which one, when.

The speed rush. The **speed rush** is the move everything else is built on: beat the tackle to the corner before he can get in front. Fast get-off, a curved path up the arc, a dip of the near shoulder as the tackle reaches for him, and a bend back toward the quarterback. It beats a tackle who sets too slowly or not deep enough, and it is the move tackles fear most, which is exactly why it makes every other move work: a tackle who is worried about speed retreats fast, and a man retreating fast is off balance in other

ways. Its weakness is the one Pass Protection built the pocket around: a tackle who gets deep enough simply runs the rusher past the quarterback.

The bull rush. The **bull rush** is the opposite idea. Instead of going around the blocker, the rusher goes *through* him: hands inside on the chest, hips low, legs churning, driving him straight back into the quarterback. A bull rush rarely produces a sack by itself; what it produces is a collapsing pocket. A quarterback who has a tackle or guard backing into his feet cannot step up, cannot finish his throwing motion and often cannot see. It beats a blocker who sets too high (pads up, so the rusher gets under him), too light, or too deep and fast, with his weight already going backward. The blocker's answer is the anchor: sink the hips, widen the base, and stop the slide. A bull rush against a blocker who anchors is a stalemate, and a stalemate is a win for the offense.

The long arm. The **long arm** is the bull rush with an escape hatch. The rusher locks one arm, fully extended, into the blocker's chest (usually his inside arm into the blocker's outside shoulder area). A long arm is longer than a blocker's punch, so the blocker cannot reach the rusher's body; the rusher controls the distance between them, drives the blocker back with it, and keeps his other hand free to knock the blocker's hands away and slip off the block when the moment comes. It has become one of the signature moves of the modern NFL edge rusher, because it works for long-armed athletes who are not giant men, and because it lets a rusher generate power without getting tied up chest to chest.

The swim. The **swim move** clears the blocker's hands with one hand and throws the other arm over the top of his shoulder, like a swimmer's overhand stroke, to get past him. On the edge it is usually an inside move: the rusher fakes outside to make the tackle widen, then swims back inside. In the middle it is a staple of quick, compact defensive tackles who meet guards almost at once. It needs timing and a blocker whose weight has gone the wrong way, and its risk is visible in the drawing: for a moment the rusher's arm is up and his chest is open, and a blocker who is not fooled will drive right into it.

The rip. The **rip move** is an uppercut: the rusher drives his inside arm up under the blocker's arm and through his armpit, lifting the arm off him and turning his own shoulder through the gap so his hips get past. It is the usual way a speed rush is *finished*, because at the corner the tackle's outside hand is the last thing between the rusher and the quarterback, and the rip takes that hand away. A dip and a rip together are the most common sight on an edge sack.

The spin. The **spin move** is the showpiece. When the tackle has committed to the outside, the rusher plants his outside foot, turns his back to the blocker and spins all the way round, coming out back inside him toward the quarterback. Dwight Freeney made it famous with the Colts in the 2000s, and it carried him to the Hall of Fame.³ Done at the right moment, against a tackle who has overset, it leaves the blocker grabbing air. Done at the wrong one, it takes the rusher out of the play entirely: for half a second he cannot see the quarterback, and a spinning rusher who meets a blocker who stayed square spins himself out of the pocket, which is one reason the move is used sparingly.

Common misconception

"Pass rushers win with strength." Strength helps, especially in the bull rush and the long arm, but most rush wins are wins of *timing and position*: a quicker first step, hands that land before the blocker's do, a move chosen for the set the blocker just showed. That mirrors the misconception from the other side of the ball in Pass Protection: most protection losses are losses of position, not of strength. The rush is a hand fight, and hands are speed and timing first.

Moves are counters to hands

Watch a pass rush closely and you will see that the rusher's hands are busy before his feet have done anything clever. The blocker's main weapon is his punch, a two-handed strike into the rusher's chest to stop his momentum and steer him. Many modern moves are built specifically to beat that punch.

The **cross-chop** is the clearest example. As the tackle punches, the rusher brings his far arm (the one away from the tackle) across his own body and chops down on the tackle's hands or forearms, knocking the punch down and away, and then rips or dips past while the tackle's hands are out of position. It turns the blocker's best tool into the reason he loses.

The **ghost (rush move)** goes one step further: no contact at all. As the tackle punches, the rusher dips his inside shoulder and turns it away, so the punch lands on nothing, and the tackle, having thrown his weight into a strike that hit air, is off balance as the rusher bends around him. A ghost works best against a tackle who punches early and hard; a patient tackle who "catches" the

³Pro Football Hall of Fame, "Dwight Freeney" (Class of 2024): <https://www.profootballhof.com/players/dwight-freeney/>; on the spin move as his trademark, Pro Football Hall of Fame, "Countdown to 2024 Pro Football Hall of Fame Enshrinement: Dwight Freeney" (August 2024): <https://profootballhof.com/news/2024/08/countdown-to-2024-pro-football-hall-of-fame-enshrinement-dwight-freeney>.

rusher and strikes late gives a ghost nothing to avoid, which is one reason, as Pass Protection noted, that many NFL tackles have stopped throwing one textbook punch at every rusher.

Rules shape this hand fight too. A rusher may use his hands on a blocker's body, but thrusting them up into the blocker's head or neck is a foul (an illegal use of hands, five yards and an automatic first down when the defense does it).⁴ That rule has a history you will meet below, and it is why the modern hand fight is aimed at the blocker's hands, arms and chest.

The rush plan: setting up the move you want

Here is the real secret of pass rushing, and the reason the moves above are only half the story. A tackle facing the same rusher sixty times in a game is not guessing blindly; he is studying the rusher too, and he adjusts. So the best rushers do not just have moves. They have a **rush plan**: a planned sequence across the game that shows the blocker one move until he adjusts to stop it, then uses the counter his adjustment has opened.

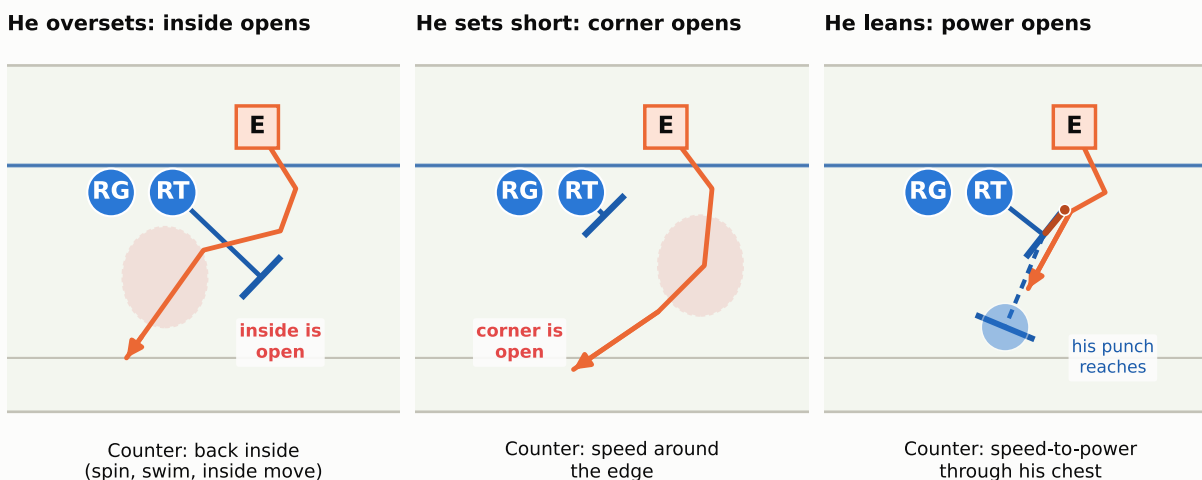


Figure 3: Why a rush plan works: every way the tackle sets opens something else (zoomed on the right tackle; the quarterback is off the bottom-left). Left: after being beaten around the edge, he kicks deep and wide at the snap to take the corner away; the space inside him (shaded) opens, and the rusher counters back inside with a spin or swim. Middle: worried about the inside, he sets short and square; now the corner is open and the speed rush wins it. Right: he leans into his punch, reaching for the rusher; the rusher turns his speed into power and drives him back (dashed), the speed-to-power. The blocker cannot protect all three at once, so the rusher's job is to make him choose and then punish the choice.

The logic in Figure 3 is a triangle. Speed threatens the corner, so the tackle must get depth. Depth leaves room inside, so the inside counter threatens that. Power threatens the tackle who reaches or leans, so he must stay balanced and patient, which slows his set, which brings speed back. A rusher with one great move can be planned for. A rusher with two that look the same for the first two steps cannot.

That is the job of **speed-to-power**, probably the single most important move in the modern edge rusher's kit. It starts exactly like a speed rush: the same get-off, the same first steps up the arc. The tackle, seeing speed, kicks back fast. Then, two or three steps in, the rusher turns straight into him and converts all that speed into a bull rush, hitting him while his weight is still going backward, when he cannot anchor. It is the move in this chapter's opening, and it works *because* of the speed rush: the more a tackle fears the corner, the faster he retreats, and the more his own momentum helps the rusher push him.

A rush plan plays out over a whole game, and you can follow it from the couch. Early, many rushers show speed again and again, to see how the tackle sets and to make him think about the corner. Once he starts bailing early, the counters come: the inside spin, the speed-to-power, the long arm. A tackle who has been beaten inside will start to sit and protect the inside, and then the speed rush is back on. Rushers and their coaches build the plan during the week from film of that specific tackle: his set depth, whether he punches early, whether he oversets after he has been beaten. The same rusher may use a completely different plan against a different tackle the following Sunday.

⁴NFL Rulebook, Rule 12, Section 1, Article 7 (use of hands by the defense): a defender may not thrust his hands or arms forward above the frame of an opponent to forcibly contact him on the head or neck; the penalty for illegal use of hands by the defense is five yards and an automatic first down. NFL Football Operations: <https://operations.nfl.com/the-rules/nfl-rulebook/>; NFL.com, "2025 rulebook, part 2": <https://www.nfl.com/news/2025-rulebook-part-2>.

i Madden vs. real life

In Madden a pass-rush move is a button press, and whether it works is mostly a matter of the two players' ratings. Real rushes don't work like that. A move works or fails depending on what the blocker just did and what he is afraid of, which depends on what the rusher showed him on the previous ten snaps. The same spin move that beats a tackle in the fourth quarter would have been stuffed in the first, before the tackle had a reason to overset. Pass rushing is a conversation across a game, not a series of independent dice rolls.

Inside rushers: less room, shorter path

Everything above is described from the edge, because that is where most pass rushers line up and most sacks are recorded. The defensive tackles play a different version of the same game.

An interior rusher has the shortest path to the quarterback, which is why Pass Protection called pressure up the middle the most destructive kind: the quarterback cannot step up into it and cannot see over it. But he also has the least room to work. A guard is on him almost immediately (there is no long arc to set up a move, only a yard or so of space), and the center is often free to help, so many of his rushes face two blockers. Interior rushers therefore win with fast, compact hands: a club or a swim the instant they meet the guard, a rip under him, a bull rush to push the pocket. And they are measured differently. A 3-technique who drives the guard back into the quarterback's lap, forcing him to slide away from the pressure and into an edge rusher's arms, may have done more than the edge rusher who gets credited with the sack.

The greatest interior rusher of the modern era made the case that a defensive tackle could be the most important player on the field. Aaron Donald, a 6-foot-1 defensive tackle the Rams drafted 13th overall in 2014, won Defensive Player of the Year three times (2017, 2018 and 2020) and had 20.5 sacks in 2018, astonishing for a player who lined up inside and was double-teamed constantly.⁵ What made him special was not size. It was get-off, hand speed and leverage: he was often past the guard before the guard had finished his first step, the one-gap penetration of Defensive Line Play applied to the pass.

Stunts: winning with geometry instead of muscle

Moves win one-on-one fights. A **stunt**, also called a **pass-rush game** or a twist, wins by making two blockers' assignments collide. In a stunt, two or more rushers exchange gaps after the snap: one goes first to occupy blockers (the **penetrator**), and the other loops around behind him into the space he left (the **looper**). The defense still rushes four, still covers with seven; it just rearranges where the four go.

You saw the most common stunt from the offense's side in Pass Protection, where the guard and tackle passed it off by switching. Here are the main families from the defense's side.

⁵“Aaron Donald,” Pro Football Reference (13th overall pick in 2014; 20.5 sacks in 2018; Defensive Player of the Year 2017, 2018, 2020): <https://www.pro-football-reference.com/players/D/DonaAa00.htm>; Associated Press, “Aaron Donald announces his retirement after a standout 10-year career with the Rams” (March 15, 2024): <https://www.ksat.com/sports/2024/03/15/aaron-donald-announces-his-retirement-after-a-standout-10-year-career-with-the-rams/>.

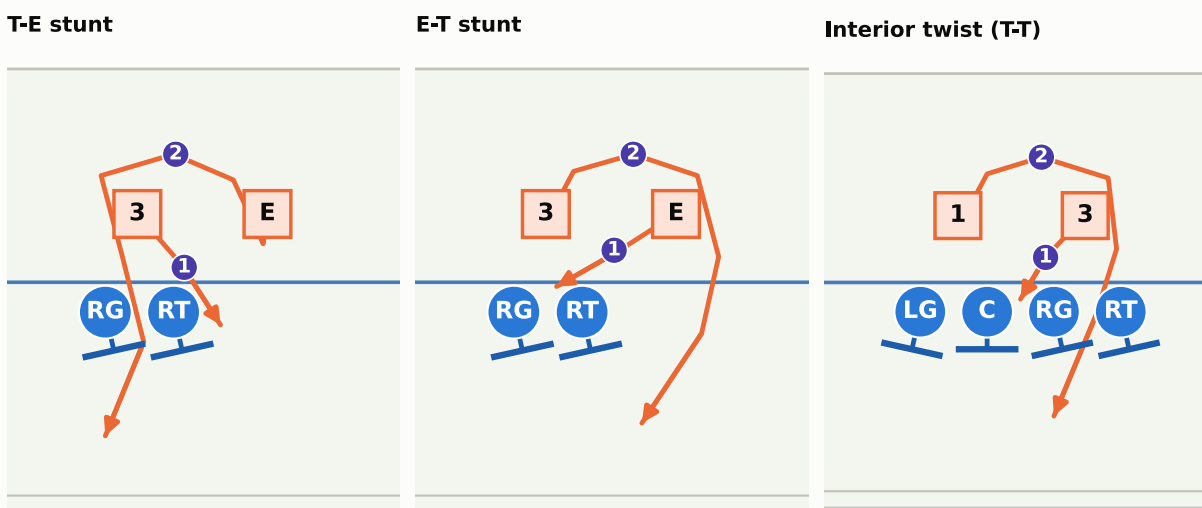


Figure 4: Three common stunts, each zoomed on the part of the line it attacks (the defensive linemen are drawn a little deeper than real, and the loops a little wide, so you can see the second man go around behind the first; the numbered badges give the order). T-E: the 3-technique goes first, crashing out into the tackle, and the end loops around behind him, on the defense's side of the line, into the gap he left. E-T: the mirror, with the end crashing inside first and the 3-technique looping around him to the outside. Interior twist (T-T): the 3-technique crashes across the center's face and the 1-technique loops behind him into the far B gap. In every case the first man's job is to occupy blockers, often two of them; the second man's job is to arrive where nobody is looking.

The T-E stunt. In a **T-E stunt** (tackle first, end second) the defensive tackle crashes outward into the offensive tackle and the end loops around behind him into the inside gap he vacated. It attacks the seam between the guard and the tackle. Its point is that a guard who follows the 3-technique outside, and a tackle who waits for the end on the outside, both end up blocking the wrong man, and the end walks through the B gap.

The E-T stunt. The **E-T stunt** is the mirror: the end crashes inside first, and the tackle loops around him to the outside, taking over the end's rush lane. It aims the penetrator at the B gap and sends the looper wide, toward a tackle who has just watched the end leave and may have turned inside to chase him.

The interior twist. Two defensive tackles can do the same thing in the middle, a **T-T twist**: one crashes across the center's face, the other loops behind him. It attacks the guards and center, the three men who are supposed to hold firm, and it is especially effective against centers who have to make the protection calls and then find a man.

Coaches use the words loosely. "Twist," "game" and "stunt" all mean the same thing in most rooms, and "loop" usually describes what the second man does. Playbooks give each stunt a code word ("Tex" is a common one for T-E), and many defenses run three-man games too, with a linebacker or a third lineman joining in.

Why stunts work, and what they cost

Stunts exist because they create a mismatch or a free rusher *without blitzing*. Against man protection, the blockers have to recognise the stunt and pass it off in a split second; if either one hesitates, the looper is free. Against slide protection, as you are about to see, a stunt aimed the wrong way is simply absorbed, but one aimed at the right spot can still choose who blocks whom.

The animation shows a T-E against a slide going the other way.

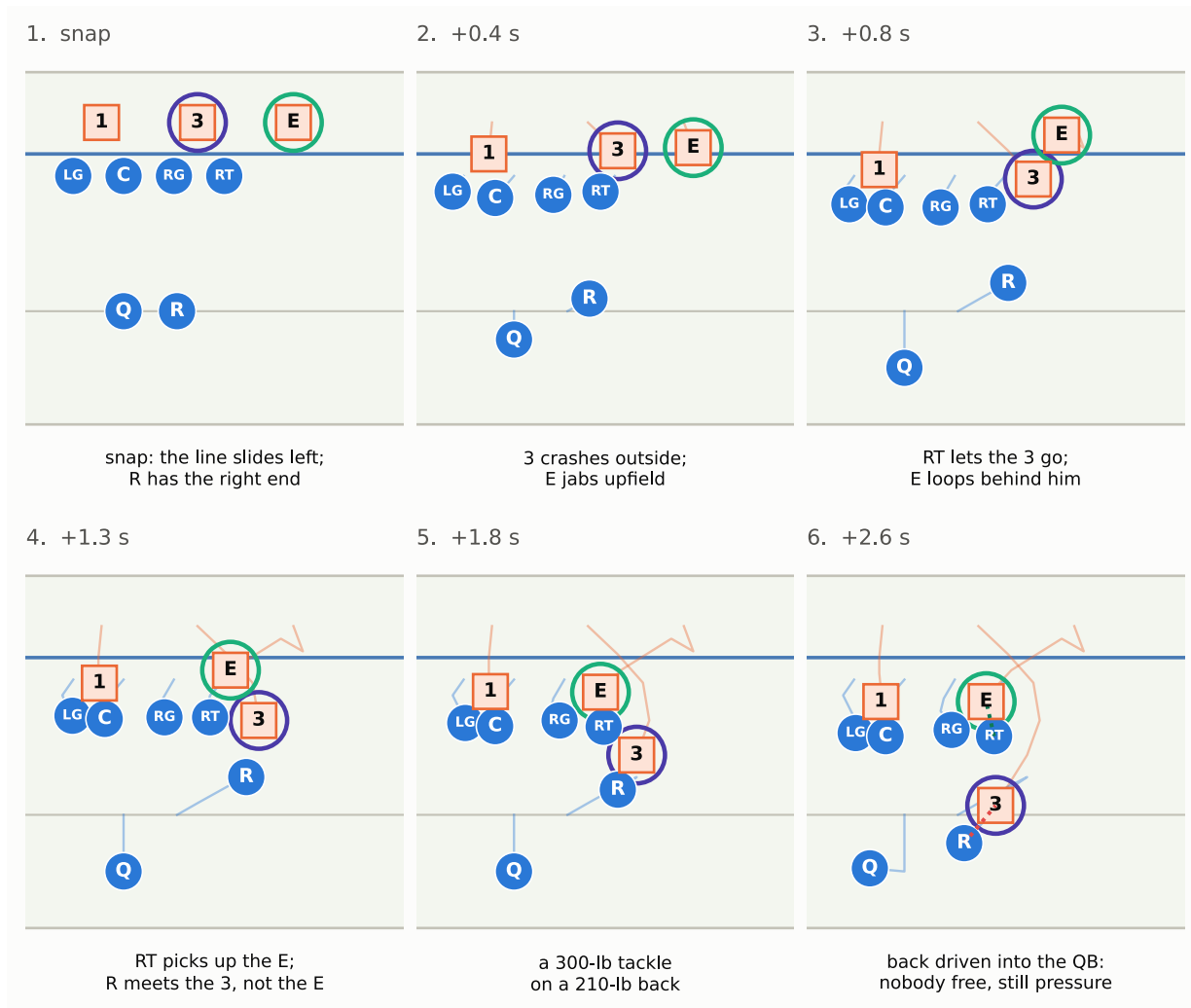


Figure 5: A T-E stunt against slide protection (simulated; zoomed on the offense's right side, linebackers and coverage not shown because they drop). The line slides left, so each lineman owns the gap to his left; the right tackle owns the B gap, and the back's job is the right end. The 3-technique (ringed in purple) crashes outside, out of the tackle's gap and into the back's area; the end (ringed in green) jabs upfield and loops behind him into the B gap. The slide does its job: the tackle keeps his gap and picks up the looping end, so nobody is free. But the stunt has changed the matchup: the back, set to block a 250-pound end, is now meeting a 300-pound defensive tackle at full speed, and he is driven back into the quarterback. Dotted lines in the last frame: who finished on whom.

Notice what the stunt did and did not do in Figure 5. It did not produce a free rusher: the slide protection handled it exactly as designed, because the right tackle owned a *gap*, not a man, and simply took whoever came into it. That is the strength Pass Protection gave slide protection: gaps do not move. What the stunt did instead was **choose the matchup**. The protection had assigned the back to the end. The stunt swapped the end for a defensive tackle who outweighs the back by ninety pounds, and the back lost. No one was unblocked, and the quarterback was still pressured.

Run the same T-E into the *slide side*, where the center, guard and tackle are all stepping toward it, and it dies: the guard owns the B gap the end loops into, the tackle owns the C gap the 3-technique crashes into, and each blocker simply takes the man who arrives. So defenses aim stunts away from the slide, at the man side of a half-slide, or at a specific blocker: a slow-footed guard, a rookie tackle, a back who has to take a lineman. Line coaches talk about stunts the way offensive coordinators talk about plays, as weapons chosen for a particular opponent.

Stunts have three costs, and they are why defenses do not stunt on every snap.

- **They take time.** The looper runs a longer path than a straight rush, so a stunt usually arrives later, about two and a half seconds or more after the snap. Against a quick throw, a stunt is a defense that rushed two men late and two men sideways. Stunts belong on long-yardage downs, when the quarterback must hold the ball for routes to develop.

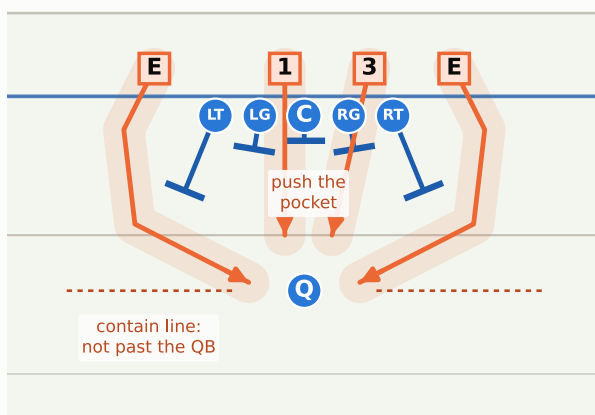
- **They give up the one-on-one.** The penetrator's job is to occupy blockers, not to win, so a great rusher who is asked to penetrate on a stunt is a great rusher not rushing. Defenses often build stunts so their best player is the looper.
- **They open lanes.** Two rushers crossing paths leave, for a moment, a space where neither of them is, and a quarterback who can run will find it. That is the next subject.

Rush lanes and contain

A four-man rush is not four men running at the quarterback. It is four men each responsible for a **lane**, and the lanes together form a box around him.

A **rush lane** is the path to the quarterback that one rusher owns. The two ends usually own the outside lanes and have contain: they rush outside the tackles but must not run past the quarterback's depth, because the space behind them would become a running lane. The two interior rushers own the middle and have the job of pushing the pocket straight back, so the quarterback cannot step up. If every rusher stays in his lane, the quarterback has nowhere to go: forward is the interior push, sideways is the edge rushers, backward is a longer throw. If one rusher leaves his lane, there is a hole.

Lanes kept: the QB is boxed in



One lane lost

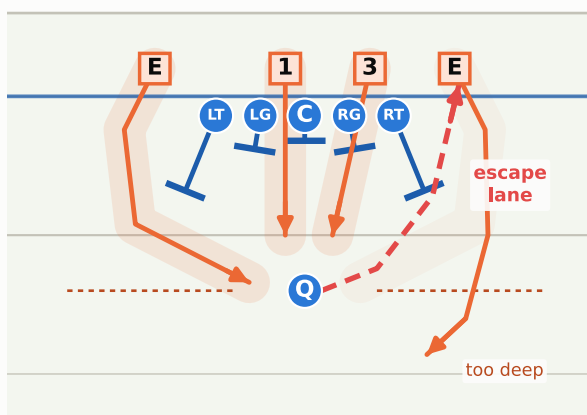


Figure 6: Rush lanes (shaded) for a four-man rush, with the quarterback at his launch point. Left: lanes kept. The ends rush outside the tackles and stop at the quarterback's depth (the dashed contain line), the inside rushers push the pocket straight back, and the quarterback is boxed in. Right: one lane lost. The right end, chasing the sack, runs his arc past the quarterback's depth; the space between him and the 3-technique becomes an escape lane (red), and a quarterback who can run steps up and out through it, often past the line of scrimmage before anyone can turn around.

The trade in Figure 6 is the oldest one in pass rushing. A rusher who stays disciplined in his lane gives the quarterback nowhere to go, but he also gives up some of his own chance to get home: the end who stops at the quarterback's depth is not running the full, fast arc that produces sacks. A rusher who abandons his lane for the sack, whether by running too far upfield, spinning inside out of an outside lane, or looping on a stunt, may get there, or may open the door. Against a quarterback who will stand in the pocket, the gamble usually pays. Against one who runs, it often doesn't.

Stunts make the lanes move. On a T-E, the penetrator crashing outside takes over the outside lane and with it contain, while the end who usually has contain loops inside. If the penetrator does not get there, or gets there too deep, the edge is open. That is part of why stunts are rarer against running quarterbacks, and why the stunts defenses do run against them are often the ones that keep a fast player outside.

The mush rush: squeezing instead of racing

Against a quarterback who is a real threat to run, defenses often go further and change what the rush is *for*. In a **mush rush**, the four rushers stay in their lanes, the ends stop level with the quarterback instead of trying to run past him, and the pocket is squeezed from all sides rather than raced into. The goal is no longer to sack him quickly. It is to make sure that when the routes are covered and he looks for somewhere to run, there is nowhere to go.

The mush rush: a cage, not a race

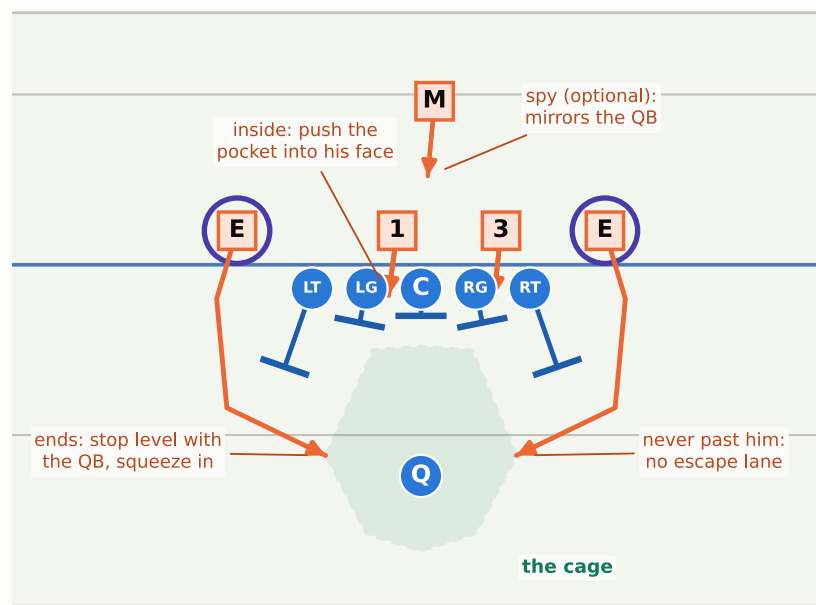


Figure 7: The mush rush against a quarterback who runs. The ends (ringed) rush up to the quarterback's depth and then squeeze inward instead of continuing up the arc; the inside rushers push straight back into his face. Together they form a cage (green outline) that shrinks around him but never opens. Some defenses add a spy, a linebacker who stays at home and mirrors the quarterback (M, here), as a last line if he does escape. The price is time: nobody in this rush is trying to beat his man quickly, so the quarterback gets longer to throw and the coverage has to hold longer.

The word “mush” sounds passive, and the mush rush is often criticised as timid when it fails. But watch it work against a great runner and it is anything but. The rushers are still fighting their blockers; they are just fighting them for *position* rather than for a path around them. An end in a mush rush may bull-rush or long-arm his tackle straight back, collapsing the pocket from the side, without ever trying to beat him around the corner. The inside rushers push the middle of the pocket back so the quarterback cannot step up. The quarterback ends up throwing from a shrinking space with no lanes, or running into a defender who was waiting for exactly that.

The cost is the clock. A mush rush produces fewer quick pressures and fewer sacks, so the quarterback gets more time and the coverage behind it has to hold for four or five seconds instead of two and a half. Against a quarterback who is both a great runner and a patient passer, that is a miserable choice, which is why such quarterbacks are so hard to defend. Some defenses add a defender (often a linebacker) assigned only to mirror the quarterback and tackle him if he runs; it buys insurance at the cost of a man in coverage or in the rush. That defender is called a spy, and Third Down has the full picture.

Common misconception

“A mush rush means the defense isn’t trying to get to the quarterback.” It means the defense has decided *where* it wants the quarterback to be, and has accepted that he will get there later. Against a quarterback who runs for a first down every time a lane opens, a sack is less valuable than an escape is costly. A mush rush that produces no sacks and no scrambles, and forces the quarterback to throw from a crowded pocket into tight coverage, is a defense winning.

Measuring a pass rush: why pressure beats sacks

A sack, the quarterback tackled behind the line before he can throw, is the pass rush’s famous statistic, and for good reason: it is one of the most damaging plays in football. It costs yards, a down and often the drive, and sometimes the ball. The NFL has counted sacks officially only since 1982;⁶ the single-season record is the 23.0 that Myles Garrett set in 2025, breaking the 22.5 shared by

⁶Sacks have been an official NFL individual statistic since 1982; earlier totals are unofficial reconstructions. “Quarterback sack,” Wikipedia: https://en.wikipedia.org/wiki/Quarterback_sack; Pro Football Reference glossary: <https://www.pro-football-reference.com/about/glossary.htm>.

Michael Strahan (2001) and T.J. Watt (2021).⁷

But as a way to *judge* a pass rush, sacks are a poor instrument, and the reason is simple arithmetic. A sack happens only when three things go right for the defense: a rusher gets to the quarterback (pressure), the quarterback fails to get rid of the ball or escape, and the rusher finishes the tackle. Only the first is mostly about the rush. So analysts split it in two:

$$\text{sacks} = \text{pressures} \times \text{the share of pressures that become sacks}$$

Pressure, as Pass Protection defined it, is any rusher forcing the quarterback to change what he wanted to do: a sack, a hit or a hurry. In FTN's charting of the 2023–2025 regular seasons, quarterbacks were pressured on 30% of dropbacks and sacked on 6.6%, so only about 22% of pressures, roughly one in 5, became sacks. The question is what decides that conversion, and whether it belongs to the rush at all.

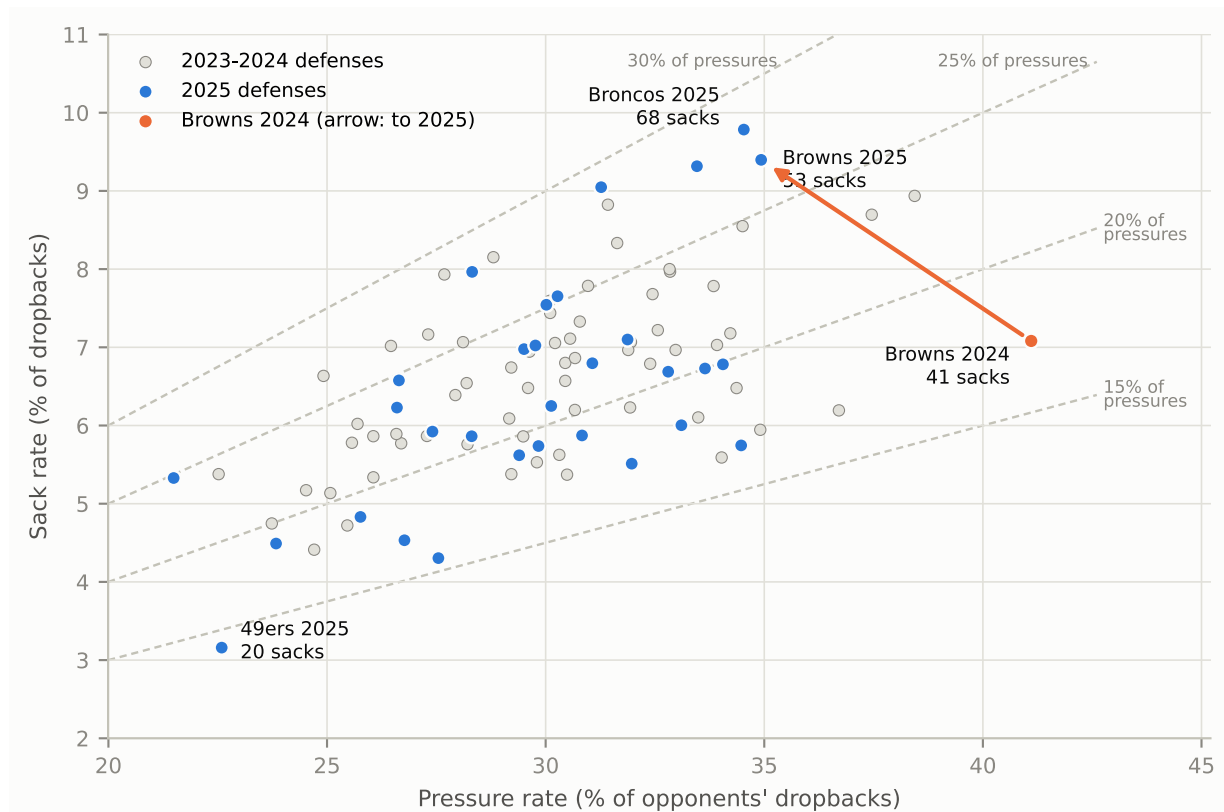


Figure 8: Pressure and sacks, defense by defense: every team-season from 2023 to 2025 (regular season; pressure includes sacks, from FTN's charting). Teams that pressure more do sack more, but look at the spread: at any pressure rate, sack rates differ by a factor of two. The faint rays show the conversion, the share of pressures that became sacks. The arrow follows the Cleveland Browns: in 2024 they had the highest pressure rate of any defense in these three seasons and an ordinary sack rate; in 2025 they pressured less often, converted far more of those pressures, and Myles Garrett set the single-season record. Data: nflverse play-by-play and participation via nflreadpy.

Three things stand out in Figure 8.

Pressure and sacks are related, but loosely. Across these team-seasons the correlation between a defense's pressure rate and its sack rate is 0.59: real, but far from one. Two defenses that reach the quarterback equally often can finish with very different sack totals. The 2025 Broncos turned 28% of their pressures into sacks, 68 in all; the 2025 49ers turned 14% into sacks and finished with 20.

The best pressure team was not the best sack team. The 2024 Browns pressured opposing quarterbacks on 41.1% of dropbacks, the highest rate of any defense in these three seasons, and finished with 41 sacks, converting only 17%. In 2025, with largely the same star at the heart of the rush, they pressured *less* often (34.9%) and finished with 53 sacks, because 27% of their pressures

⁷NFL.com, "Browns DE Myles Garrett reaches 23 sacks, sets new NFL single-season record" (Week 18, 2025, against Cincinnati; the previous record of 22.5 was shared by Michael Strahan, 2001, and T.J. Watt, 2021): <https://www.nfl.com/news/browns-de-myles-garrett-reaches-23-sacks-sets-new-nfl-single-season-record>.

became sacks. Garrett's record season was real, and so was his dominance the year before; the sacks just arrived in one year rather than the other.

So which number should you believe? The test is which one *repeats*. A measure of skill should carry over from one season to the next, because the players and coaches mostly carry over; a measure full of luck should not.

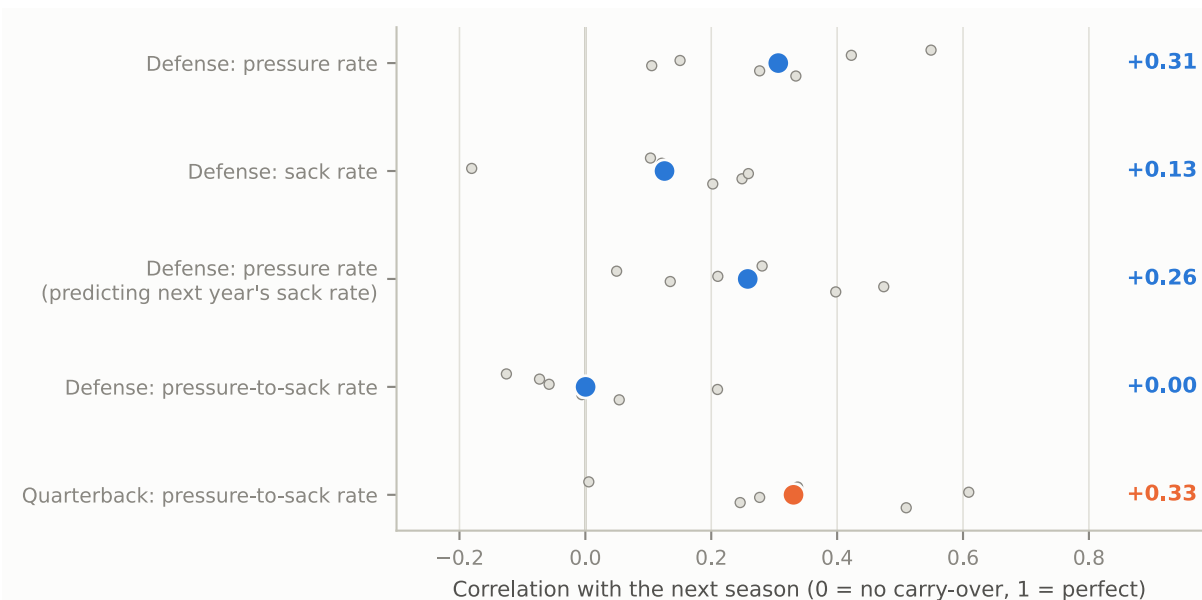


Figure 9: What carries over from one season to the next. Each row is a correlation between a measure in one season and the same team's (or quarterback's) measure the next season; the faint dots are the individual season pairs (2018-19 to 2021-22 from Next Gen Stats' charting, 2023-24 and 2024-25 from FTN's; the 2022-23 pair is left out because the source changes), and the large dot is their average. A defense's pressure rate repeats far better than its sack rate, and even predicts next year's sacks better than this year's sacks do. A defense's pressure-to-sack conversion does not repeat at all; a quarterback's does. Data: nflverse play-by-play and participation via nflreadpy; quarterbacks with at least 100 pressured dropbacks in both seasons.

Figure 9 is the core of the case for pressure.⁸ On average, a defense's pressure rate correlates about 0.31 with its own pressure rate the next season; its sack rate correlates only about 0.13 with next season's sack rate. And this season's *pressure* rate predicts next season's *sack* rate (about 0.26) better than this season's sack rate does. If you want to know which pass rush will get sacks next year, count its pressures this year, not its sacks.

The fourth row shows why. A defense's conversion rate, the share of its pressures that become sacks, has essentially no carry-over (an average correlation of +0.00). A defense that converted a lot one year is no more likely than any other to convert a lot the next. Whatever drives the conversion, it is not something the defense keeps.

The fifth row says what it is, or at least who. A *quarterback's* conversion rate, the share of the times he is pressured that end in a sack, does carry over, about 0.33 on average (noisier, because there are only twenty or thirty quarterbacks a season with enough pressures to measure). Some quarterbacks get rid of the ball, throw it away, step up or escape when the rush arrives, and some take the sack, year after year. Quarterback Play measured how wide that spread is; this is why it matters for the rush. A pass rush's sack total depends a great deal on which quarterbacks it happened to face, how often it faced them on third-and-long, and a large helping of luck about which pressures arrived a half-step before the throw. Pressure is the part of the sack the rush owns.

⁸Author's calculation via nflreadpy: nflverse play-by-play (qb_dropback, sack) joined to the participation file's was_pressure, regular seasons 2018–2025. Pressure includes sacks (Next Gen Stats, which supplies 2018–2022, leaves the flag blank on sacks; FTN, 2023–2025, fills it). Correlations are Pearson, across the 32 defenses (or across quarterbacks with at least 100 pressured dropbacks in both seasons) for each consecutive pair of seasons within one source: 2018–19, 2019–20, 2020–21, 2021–22, 2023–24, 2024–25. Participation data dictionary: https://nflreadr.nflverse.com/articles/dictionary_participation.html.

Common misconception

“Sacks tell you who the best pass rushers are.” They tell you who *finished* plays, which is real but partial. Many sacks are **clean-up sacks**: an interior rusher collapses the pocket, the quarterback slides away from him into an edge rusher, and the edge rusher gets the credit. Others are **coverage sacks**, where the quarterback held the ball for four or five seconds because nobody was open, and whoever arrived first got the stat. And some of the best rushes in a game, like the one that opened this chapter, produce a throw a beat too early and no statistic at all. Watch for who *caused* the pressure, not just who was standing over the quarterback.

Pass rush win rate, get-off and time to pressure

If sacks are too noisy and pressure is better, can we do better still? Pressure has its own flaw: it depends on the clock. A rusher who beats his man cleanly in 1.5 seconds against a quarterback who throws in 1.3 gets no pressure, and a rusher who is stonewalled for four seconds and then wanders in while the quarterback pats the ball gets one. Tracking data allows measures that do not depend on what the quarterback does.

Pass rush win rate. ESPN’s **pass rush win rate (PRWR)**, introduced in 2018 by ESPN Analytics and built on Next Gen Stats tracking data, asks one question of every rush: did the rusher beat his block within 2.5 seconds of the snap?⁹ Its companion, pass block win rate, asks the reverse of every blocker. The fixed 2.5-second clock is the clever part. It sits a little before the average throw, ESPN found it best separated the rushers and blockers already known to be good, and it takes the quarterback out of the number: a quarterback who holds the ball cannot hand rushers late “wins,” and one who throws quickly cannot hide a rusher who was about to get home. ESPN publishes it as leaderboards ranked within position groups, so edge rushers are compared with edge rushers and interior linemen with interior linemen, because their jobs are so different.

Its limits are worth knowing. A “win” is not a pressure; a rusher can beat his man and still be too late or too wide to matter. The model does not know each rusher’s assignment, so a penetrator on a stunt, whose job is to occupy blockers, can look like a loser while doing exactly what was asked. And the play-by-play data is not public, only the leaderboards, so you cannot rebuild it yourself. Use it the way you would use a scout’s opinion: an excellent tiebreaker between players, not a final verdict.

Get-off and time to pressure. Two simpler tracking measures bracket the rush. Get-off, as above, is how quickly a rusher crosses the line after the snap. **Time to pressure** is the seconds from the snap to the first pressure on the play. Together they separate the rushes that won from the ones that were handed something: a pressure inside about two and a half seconds is almost always a rush win; one after three and a half is usually the coverage, the quarterback, or both. Here is how each is computed from tracking data, on a simulated four-man rush.

⁹Brian Burke, “We created better pass-rusher and pass-blocker stats: How they work,” ESPN, October 5, 2018: https://www.espn.com/nfl/story/_/id/24892208/creating-better-nfl-pass-blocking-pass-rushing-stats-analytics-explainer-faq-how-work.

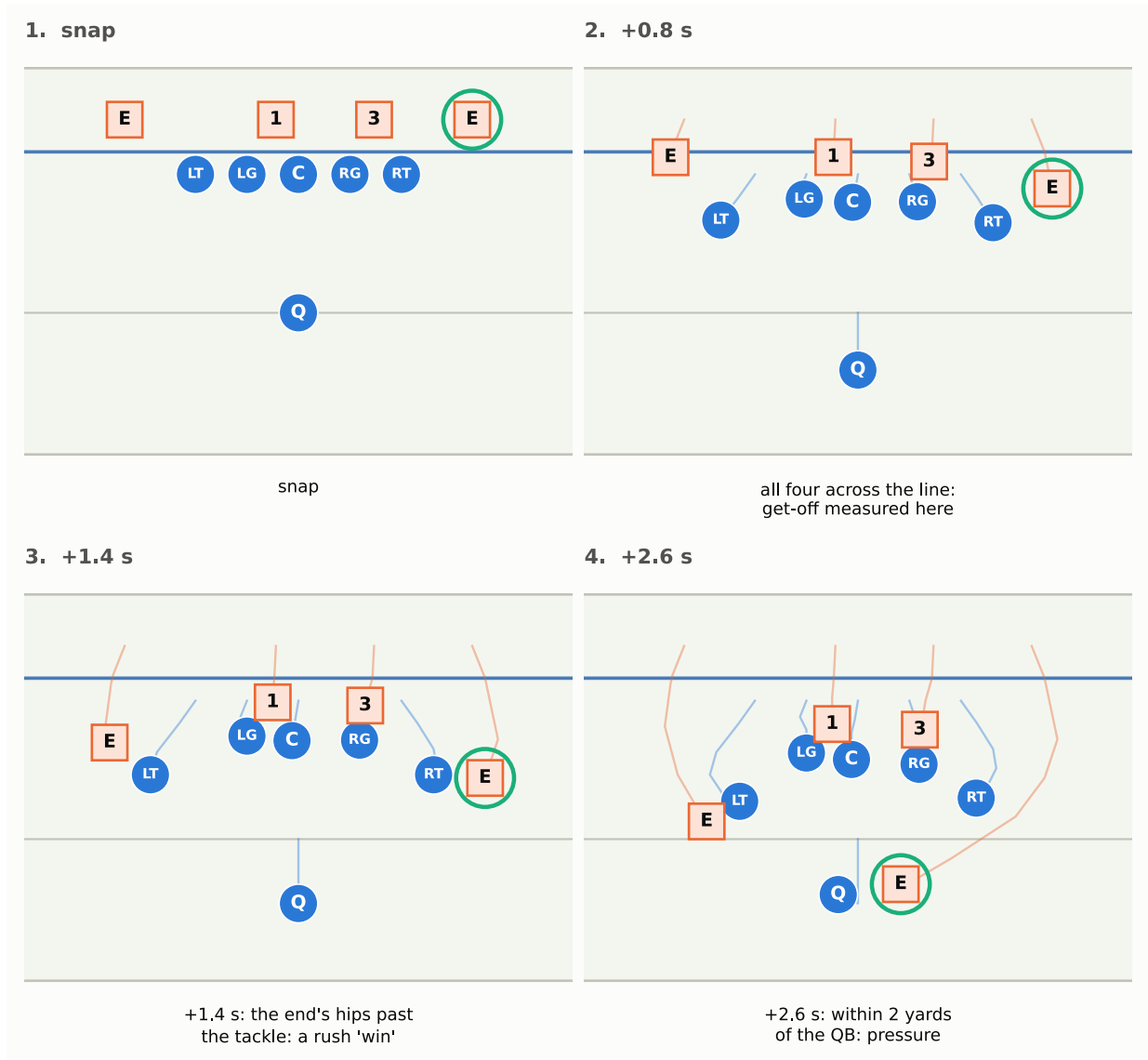


Figure 10: A four-man rush measured from tracking data (simulated, BDB-format; receivers and coverage not shown). 1: the snap. 2: well under a second later all four rushers are across the line; each one's get-off is the moment he crossed it. 3: the right end's hips get past the right tackle, the moment his rush is won, well inside the 2.5-second window pass rush win rate uses. 4: the end reaches two yards of the quarterback, this chapter's working definition of pressure for the measurement, and the quarterback has to move.



Figure 11: The same simulated rush as numbers: each rusher's distance from the quarterback, frame by frame (ten per second). The dots mark each rusher's get-off, the frame he crossed the line. The dashed horizontal line is the two-yard pressure threshold used here; the right end crosses it first, which is the play's time to pressure. The shaded band is pass rush win rate's 2.5-second window: the end beat his blocker inside it, so his rush counts as a win, while the other three never beat theirs.

In this simulated play the rushers' get-offs range from 0.5 to 0.8 seconds, the right end beats his tackle at 1.4 seconds, and the time to pressure is 2.6 seconds. On a real play the definitions need more care than a two-yard circle (data providers use their own rules for what counts as pressure, and real players are not points), but the logic is exactly this, and it is the logic you will build on real Big Data Bowl frames in Tracking Data and the Big Data Bowl.

Where the moves came from

The pass rush as a craft, with named moves, rush plans and a hand fight, grew out of a series of rule changes, each of which took a weapon away from one side or gave one to the other.

The modern pass rusher's ancestor is Deacon Jones, the Los Angeles Rams' defensive end of the 1960s, a ferocious speed rusher who is credited with coining the word "sack" for what he did to quarterbacks. His signature weapon was the **head slap**: a blow to the side of the blocker's helmet at the snap to stun him for an instant, then go around him. The NFL banned the head slap in 1977, and blows to the head have been illegal for rushers ever since.¹⁰ A year later the 1978 rule changes went the other way for blockers, letting them extend their arms and open their hands on pass blocks (Pass Protection and The 1978 Liberation tell that story). Together, the two changes turned the pass rush into what it is now: a fight between the rusher's hands and the blocker's, aimed at arms and chests, where technique and timing matter as much as violence.

Then came the player who made offenses rebuild their protections around him. Lawrence Taylor, the Giants' outside linebacker from 1981, rushed from a two-point stance on the edge of a 3-4 defense with a speed no blocker of the time could match. Teams that left a running back to block him lost; Joe Gibbs's Washington offense famously answered with an extra tight end and an H-back to put more bodies on him (the story is in Pass Protection). In 1986 Taylor had 20.5 sacks and was the league's Most Valuable Player, the last defensive player to win the award.¹¹ The value of the left tackle, the man who blocks a right-handed quarterback's blind side, rose with him.

Reggie White, "the Minister of Defense," showed what power and hands could do. After fifteen seasons, most of them with the Eagles and the Packers, he retired as the NFL's career sack leader, with 198 (Bruce Smith later passed him, finishing with 200), and his signature, the "hump" move, was a forearm club up under the blocker's arm that lifted and threw him aside: a rip and a bull

¹⁰"Deacon Jones," Wikipedia (credited with coining "sack"; the head slap, banned in 1977): https://en.wikipedia.org/wiki/Deacon_Jones; Pro Football Hall of Fame, "Deacon Jones": <https://www.profootballhof.com/players/deacon-jones/>. Who first used "sack" is commonly attributed to Jones; the attribution rests on his own account.

¹¹"Lawrence Taylor," Wikipedia (1986: 20.5 sacks, NFL MVP; the most recent defensive player to win the AP award): https://en.wikipedia.org/wiki/Lawrence_Taylor; Pro Football Hall of Fame, "Lawrence Taylor": <https://www.profootballhof.com/players/lawrence-taylor/>.

¹²Pro Football Hall of Fame, "Reggie White": <https://www.profootballhof.com/players/reggie-white/>; "Reggie White," Wikipedia (198 career sacks, the record at his retirement; the "hump" move): https://en.wikipedia.org/wiki/Reggie_White; Bruce Smith's 200 career sacks: "Bruce Smith," Wikipedia, https://en.wikipedia.org/wiki/Bruce_Smith.

rush in one violent motion.¹² In the 2000s Dwight Freeney’s spin move made a smaller, quicker body type dangerous on the edge. And in the 2010s Aaron Donald proved that the most dangerous rusher on the field could line up inside.

Today’s edge rushers are the products of all of it, and more. They study tackles’ sets on film like quarterbacks study coverages, many work with private pass-rush coaches in the offseason, and the hand fight has become its own discipline: the long arm, the cross-chop, the ghost. T.J. Watt tied the sack record in 2021 and won Defensive Player of the Year; Myles Garrett broke it in 2025 and won the award unanimously.¹³ And Micah Parsons, the Cowboys’ first-round pick in 2021, made a different point: drafted as an off-ball linebacker, he became one of the league’s most feared rushers by lining up all over the front, standing up over a guard one snap and wide on the edge the next, so that protections could never be sure where to slide. Dallas traded him to Green Bay just before the 2025 season, for two first-round picks and defensive tackle Kenny Clark.¹⁴

Film room

i Film room: Cleveland Browns 2025 — Myles Garrett’s record

The season. Myles Garrett set the single-season sack record with 23.0 in 2025, breaking it with a Week 18 sack against Cincinnati, and was voted Defensive Player of the Year unanimously. As the data section showed, the Browns as a team *pressured* quarterbacks more often in 2024; 2025 was the year the pressures turned into sacks.

What to notice: pick any Browns passing down from 2025 and watch only Garrett and the man blocking him.

- **The get-off.** Count how often he is the first player on the field to move. Watch where he meets the tackle: in front of him, or level with him at the corner.
- **The first move, then the second.** Note what he shows in the first quarter (speed up the arc?) and what he does in the fourth. Look for the speed-to-power: three steps of speed, then straight into the tackle’s chest.
- **The help.** Count how often the offense chips him with a back or tight end, or slides its protection toward him. Every extra man on Garrett is a one-on-one for someone else.
- **The finish.** On his sacks, notice how many came quickly (a rush win) and how many came after the quarterback had held the ball (a coverage or clean-up sack).

i Film room: Super Bowls LIX and LX — pressure with four

The games. The last two Super Bowls were won by defenses that got to the quarterback without blitzing. In Super Bowl LIX (February 2025) the Eagles rushed four or fewer on 44 of Patrick Mahomes’s 45 dropbacks and sacked him 6 times, 6 of them with four or fewer rushers. In Super Bowl LX (February 2026) the Seahawks rushed four or fewer on 46 of Drake Maye’s 54 dropbacks and sacked him 6 times, 5 of them with four or fewer.¹⁵

What to notice:

- **Count the rushers on every dropback.** Almost always four. Then count the coverage behind them: seven men for at most five receivers.
- **Look for the stunts.** On long downs, watch for two rushers crossing paths, and ask which blocker the looper was aimed at.
- **Time the pressure.** Count from the snap to the moment the quarterback has to move. Most of the damaging pressures in both games came early enough that no receiver had finished his route.

The Eagles’ game is studied in the Eagles case study, and Seattle’s defense, built by Mike Macdonald, in Fangio’s Two-High Revolution and Macdonald’s Answer.

¹³Garrett’s unanimous Defensive Player of the Year award for 2025: NFL.com, “List of NFL Honors award winners from 2025 NFL season,” <https://amp.nfl.com/news/list-of-nfl-honors-award-winners-from-2025-nfl-season>. Watt’s 22.5 sacks and Defensive Player of the Year award in 2021: “T. J. Watt,” Wikipedia, https://en.wikipedia.org/wiki/T._J._Watt.

¹⁴“Micah Parsons,” Wikipedia (12th overall pick by Dallas in 2021; Defensive Rookie of the Year; traded to the Green Bay Packers on August 28, 2025, for Kenny Clark and first-round picks in 2026 and 2027): https://en.wikipedia.org/wiki/Micah_Parsons; ESPN, “Sources: Micah Parsons traded to Packers, gets record contract”: https://www.espn.com/nfl/story/_/id/46098144/sources-micah-parsons-traded-packers-gets-record-contract.

¹⁵Author’s calculation from the nflverse participation file (FTN charting: `number_of_pass_rushers`, `was_pressure`) joined to play-by-play, games 2024_22_KC_PHI and 2025_22_SEA_NE. The six sacks in each game match published accounts: CBS News on Super Bowl LIX (Eagles 40, Chiefs 22; MVP Jalen Hurts), <https://www.cbsnews.com/amp/news/nfl-mvp-super-bowl-jalen-hurts-eagles>, and “Super Bowl LX,” Wikipedia (Seahawks 29, Patriots 13; Drake Maye sacked six times), https://en.wikipedia.org/wiki/Super_Bowl_LX.

Watch for it on Sunday

Watch for it

Before the snap.

- **Find the best rusher and his alignment.** Wide outside the tackle (built for speed), tight over him (power and inside moves), or standing up over a guard (he is being moved around to beat the protection).
- **Count the rushers you expect and the blockers you see.** Four rushers against five or six blockers means the defense is trusting its four; look at the coverage behind them.
- **Check the down and distance.** Third-and-long is when stunts and patient rushes have time to work; on third-and-2 the ball will be out too fast for either.
- **Is the quarterback a runner?** If so, expect the ends to rush with discipline, and maybe a spy.

After the snap: the stopwatch drill.

- **Count the seconds** from the snap to the first pressure (“one-Mississippi”), or use a stopwatch on a replay. Under about two and a half seconds is a rush win. Past three and a half, the coverage or the quarterback owns most of it.
- **Note whether the ball was already out.** A rusher who arrives just after the throw has still done his job if he made the quarterback throw early; a rusher who arrives after a five-second hold was handed it.
- **Name the move.** Around the corner (speed and rip), through the chest (bull or long arm), or back inside (swim, spin). In the fourth quarter, look for the counter to whatever worked in the first.
- **Watch for crossing paths.** Two rushers swapping lanes is a stunt; find the looper, and see which blocker he was aimed at, and whether the blockers passed it off.
- **Check the lanes.** If the quarterback escapes, freeze the replay and find the rusher who left his lane.

For your prediction log: for one quarter, write down on every dropback the number of rushers, your stopwatch time to pressure (or “none”), and whether the ball was out first. You will see which defenses win early and which only win late.

Predict the play

💡 Predict the play 1: third-and-long against five

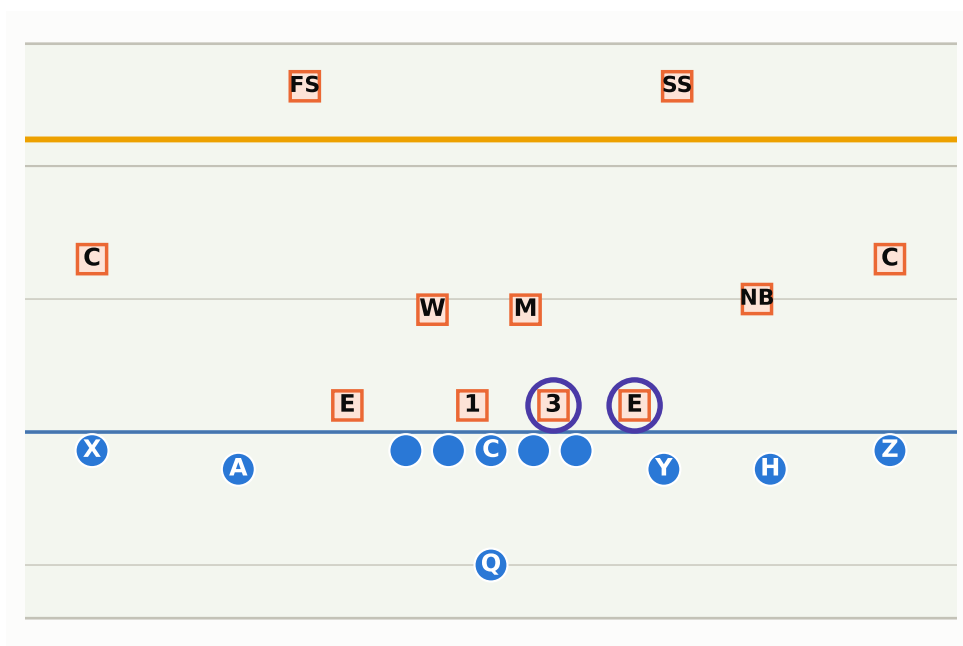


Figure 12: Third-and-11. The offense is in an empty formation, three receivers right and two left, so only the five linemen can block. The defense shows four linemen (a 1-technique, a 3-technique to the right, an end outside each tackle), two linebackers off the ball and two deep safeties. The two defenders on the right side of the line are ringed.

Is this a good down for a stunt? If so, which one, aimed where, and what is the defense risking?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the quarterback who runs

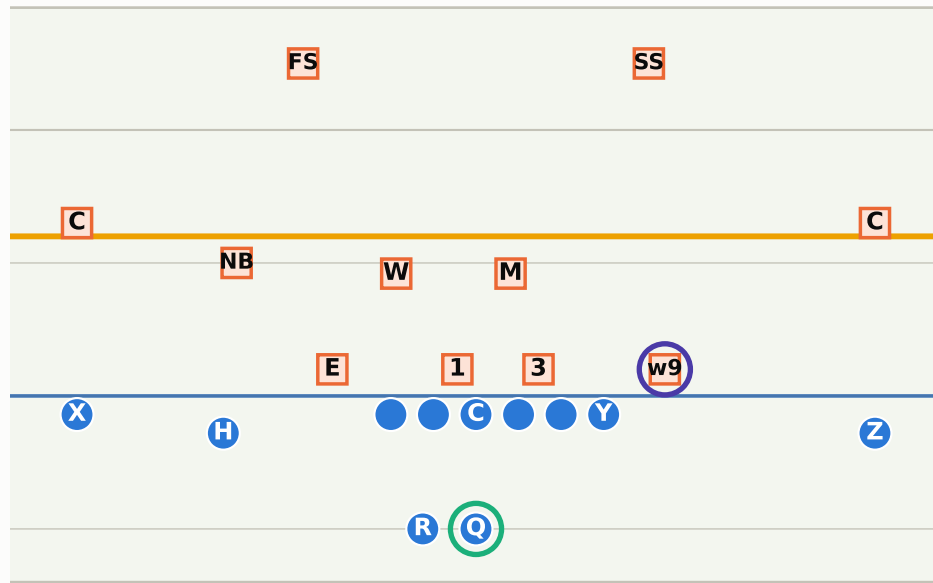


Figure 13: Third-and-6. The offense is in the shotgun with three wide receivers, a tight end, and a back beside a quarterback (ringed in green) who is one of the league's most dangerous runners. The defense is in a nickel with four linemen, including a wide-9 end on the right (ringed in purple).

How do you expect the four to rush, which of them has the hardest job, and what is the defense giving up?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the counter

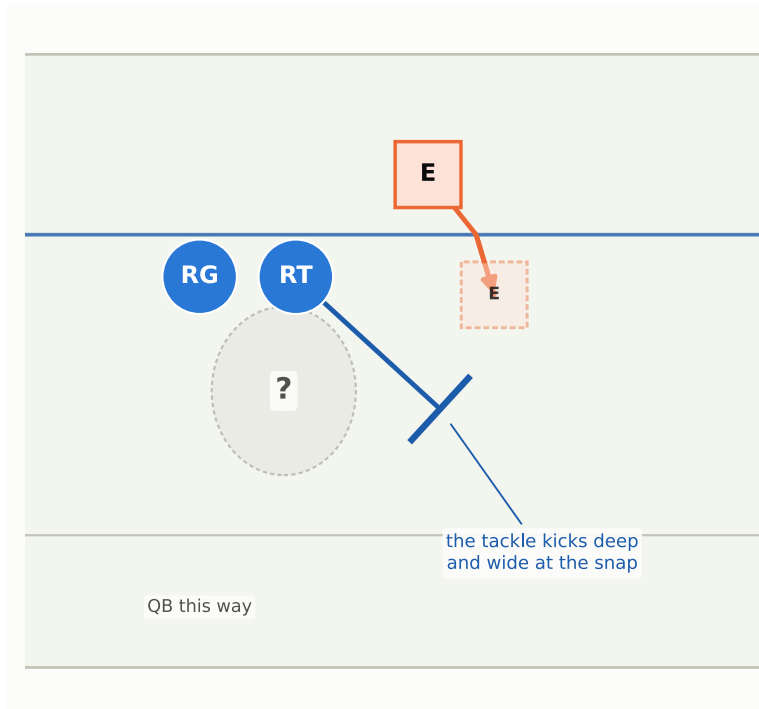


Figure 14: Fourth quarter. The defense's best rusher (E) has beaten this right tackle around the outside twice today, once for a sack. On this snap the tackle has kicked back hard and wide at the snap, to take the corner away. The picture is about 0.8 seconds after the snap: the faint square is where the rusher is now, near the top of his arc.

What move is coming, and what can the offense do about it, this snap and the next?

Answer 3 is at the end of the chapter. Try it first.

Back to the replay

Return to the play that opened the chapter. Third-and-8, four rushers, an end who took three strides up the arc and then drove straight into the tackle: **speed-to-power**, set up by every speed rush he had shown before. The tackle, fearing the corner, had kicked back fast, so his weight was going backward when the end turned into him, and he could not anchor. The pocket collapsed from the side, the quarterback threw early, and the pass sailed.

The box score has nothing for the end. FTN's charters would mark that dropback as pressured. Pass rush win rate would very likely count it as a win, his block beaten well inside 2.5 seconds. And a stopwatch would put the pressure at about two seconds, early enough that it was entirely the rush's doing. That is the whole chapter in one snap: the moves, the plan that set them up, and the reason a coach values the hurry that nobody counts more than the sack total everybody does.

Takeaways

- **The rush is a race and a fight.** A rusher must beat a bigger blocker before the ball is out, a little under three seconds on average. Get-off decides where they meet: a tenth of a second is most of a yard, and meeting the tackle level at the corner instead of in front of him is the difference between winning and losing.
- **Each move punishes a mistake.** Speed beats a slow set; the bull rush and long arm beat a high, light or retreating one; the swim, rip and spin beat a blocker who has committed. Cross-chops and ghosts beat the punch itself.
- **The best rushers have a plan.** They show one move until the tackle adjusts, then use the counter his adjustment opens; speed-to-power works *because* tackles fear speed.
- **Stunts win with geometry.** T-E, E-T and interior twists exchange rushers to beat man blocking or choose matchups against slides, at the cost of time and of lanes.
- **Four rush lanes make a box.** Ends keep contain, the middle pushes the pocket; against a quarterback who runs, the mush rush squeezes instead of racing and accepts a longer clock.

- **Count pressure, not sacks.** Pressure carries over from season to season and predicts future sacks; the share of pressures that become sacks mostly does not, except as a trait of the quarterback. Pass rush win rate, get-off and time to pressure use tracking data to take the quarterback out of the measure.

Connections

Builds on: Pass Protection (the pocket, slide and man protection, sets, punch and anchor, passing off stunts, and the pressure numbers, all seen from the other side here) and Defensive Line Play (technique numbers, one-gap penetration, the wide-9 and contain). The quarterback's share of sacks comes from Quarterback Play.

Sets up: Blitzing, where the defense stops trying to win with four and adds rushers instead, and Zone Blitzes and Simulated Pressure, the modern art of making four rushers come from places the protection did not expect; Third Down, for spies and rushing three; and Tracking Data and the Big Data Bowl, where you will measure get-off and time to pressure on real frames. The data section's metrics are mapped against the rest of football's numbers in Expected Points and Success Rate.

Case study: The Eagles: Trenches, RPOs and the Tush Push, including Super Bowl LIX's four-man pressure.

i Go deeper

- ESPN's pass rush and pass block win rate leaderboards (search "ESPN pass rush win rate"), and Next Gen Stats' pass-rushing pages, for the tracking-based numbers in this chapter.
- Pat Kirwan, *Take Your Eye Off the Ball* (2010; revised 2015): a former NFL coach and scout on how to watch the line of scrimmage, including the pass rush and protection.
- Michael Lewis, *The Blind Side* (2006): the story of how Lawrence Taylor changed what teams were willing to pay offensive tackles.
- The many pass-rush technique breakdowns by former players and position coaches on coaching sites and video channels: watch them with this chapter's vocabulary (set, arc, corner, counter) and you will see the rush plan in every one.

Answers to Predict the play

Answer 1.

Yes: this is the down stunts are made for, and the T-E on the right, with the 3-technique crashing out and the end looping inside, is the natural call. Three things point to it. *The clock:* on third-and-11 the quarterback needs routes that get past the line to gain, which take time to develop, so he will usually hold the ball two and a half seconds or more, long enough for a loop to arrive. *The numbers:* four rushers against five blockers is a fair fight, and with nobody in the backfield there is no back to absorb a mistake; every blocker is already committed to a rusher or a gap. *The geometry:* the 3-technique and the end on the same side can attack the seam between the right guard and right tackle, the classic T-E target, forcing those two to pass the stunt off cleanly in a split second.

Two cautions. If the offense's five-man protection slides toward that side, the center, right guard and right tackle all step that way and own those gaps, and the stunt is passed off automatically, which is why defenses study how a team protects in empty before calling it. And the stunt moves the right side's contain from the end to the 3-technique: if the quarterback can run and the 3-technique gets washed past him, third-and-11 becomes a scramble for a first down. Against a pocket passer on third-and-long, that is a risk most defenses happily take.

Answer 2.

Expect a mush rush: four rushers who keep their lanes, ends who stop level with the quarterback, and inside rushers who push the pocket back into his face. On third-and-6, a quarterback who runs does not need a receiver to get open; he needs one lane. So the defense's first priority is that no lane opens, and its second is the sack.

The wide-9 end has the hardest job. His alignment is built for a long, fast arc around the tackle, exactly the rush that runs an end past the quarterback and opens the escape lane in Figure 6. Against this quarterback he has to flatten his rush early, stay at the quarterback's depth and squeeze inward, and use power (a bull rush or long arm) to collapse the pocket from the side instead of trying to beat the tackle around it. Expect fewer stunts, or only ones that keep a fast player outside, and don't be surprised if a linebacker, maybe the Mike, mirrors the quarterback instead of covering anyone: a spy.

The price is time. Nobody in a mush rush is trying to win quickly, so the quarterback gets longer to throw, and the seven (or six, with a spy) defenders in coverage must hold for four seconds or more. A great runner who is also patient can wait for a receiver

to come open, which is why such quarterbacks are so hard to defend. And if anyone breaks contain, the play you will see is the one the defense feared most: a quarterback through the hole and past the line to gain.

Answer 3.

An inside counter: a spin, an inside swim or an inside rip, straight into the space the tackle has just vacated (the question mark). This is the rush plan paying off. Twice the rusher won the corner with speed; now the tackle, afraid of the corner, is bailing out deep and wide before the rusher has even committed. That leaves the gap between the tackle and the guard empty, and the shortest path to the quarterback runs right through it. A rusher who sees the tackle open his hips early plants his outside foot and comes back underneath, and a tackle moving that fast in the wrong direction cannot stop and recover. The other good answer is speed-to-power: three steps up the arc, then straight into a tackle whose weight is going backward and who cannot anchor.

The offense has several answers. **This snap:** the right guard can help, sliding out a step to fill the gap he can see opening (in a slide protection toward that side he owns it anyway). **Next snaps:** the tackle varies his sets, a jump set or a 45-degree set instead of always bailing, so the rusher cannot read the set and pick the counter; the offense chips the rusher with a back or tight end on the way out, so the tackle can set without fearing the corner; and the play caller calls quick throws, so the ball is gone before any move can finish. That is the conversation across a game in miniature: the rusher made the tackle afraid of one thing, and now the offense has to change the question.